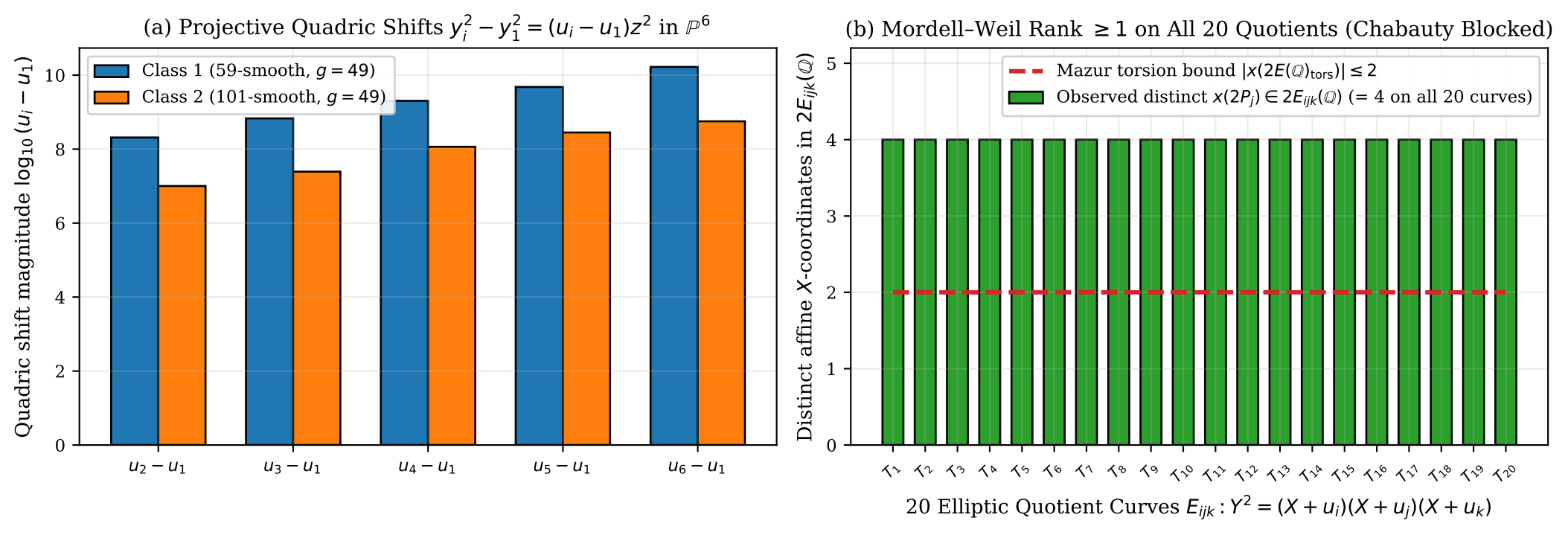
(a) Projective Quadric Shifts $y_i^2 - y_1^2 = (u_i - u_1)z^2$ in $\mathbb{P}^6$
Class 1 (59-smooth, $g = 49$)
Class 2 (101-smooth, $g = 49$)
Quadric shift magnitude $\log_{10}(u_i - u_1)$
0
2
4
6
8
10
$u_2 - u_1$
$u_3 - u_1$
$u_4 - u_1$
$u_5 - u_1$
$u_6 - u_1$
(b) Mordell–Weil Rank ≥ 1 on All 20 Quotients (Chabauty Blocked)
Mazur torsion bound $|x(2E(\mathbb{Q})_{\text{tors}})| \leq 2$
Observed distinct $x(2P_j) \in 2E_{ijk}(\mathbb{Q})$ (= 4 on all 20 curves)
Distinct affine X-coordinates in $2E_{ijk}(\mathbb{Q})$
0
1
2
3
4
5
T_1 T_2 T_3 T_4 T_5 T_6 T_7 T_8 T_9 T_{10} T_{11} T_{12} T_{13} T_{14} T_{15} T_{16} T_{17} T_{18} T_{19} T_{20}
20 Elliptic Quotient Curves $E_{ijk} : Y^2 = (X + u_i)(X + u_j)(X + u_k)$